Darwin Nuñez Chavarría

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PROFESSIONAL SUMMARY

Software Developer and Information Systems Engineering graduate with professional experience refactoring a production PHP API using an internal framework, plus hands-on experience building a full-stack web platform for a real client and a fully local RAG system with LangChain, FAISS, and Ollama. Experienced in collaborative development, legacy-code maintenance, administrative workflows, data import/export, and security-conscious implementation.

TECHNICAL SKILLS

Programming: PHP, Java, Python, JavaScript, C, C++, C#, Kotlin, SQL, Bash
Frameworks & Architecture: Flameware (internal framework), Laravel, Blade, React Native, Node.js, .NET, REST APIs, microservices, MVC, Screaming Architecture
AI & Data: RAG, LangChain, FAISS (vector databases), embeddings, information retrieval, Ollama, prompt engineering, OCR (Tesseract), PyMuPDF
Databases & DevOps: MySQL, PostgreSQL, SQL Server, Docker, Docker Compose, Linux, Git
Security & Tools: Serialization, CAPTCHA, security verification, Nmap, Metasploit, OpenCode
Cloud & Other Tools: Azure, Google Cloud, Nginx, Cloudflare, VS Code, Visual Studio, Android Studio, SQL Server Management Studio
Professional Skills: Teamwork, problem-solving, communication, adaptability, time management, leadership

PROFESSIONAL EXPERIENCE

Software Developer – API Refactoring | *Orlando Caterers And Commissary* Apr 2026 – Present

- Refactor a production PHP API using the company's internal **Flameware** framework, incorporating the framework's conventions alongside existing architectural practices (**Screaming Architecture**).
- Use **Docker** to run the development database and work with **PHP 8.5** in the Linux terminal (CachyOS).
- Apply security verification practices and supporting tools during API refactoring and implementation.
- Work with serialization and CAPTCHA-related components as part of the application's implementation and verification.

PROJECTS

Games History ChatBot – Retrieval-Augmented Generation (RAG) System | *Python, LangChain, FAISS, Ollama, Llama 3*

- Built a fully local RAG pipeline over a 1,141-page corpus of five primary sources, indexing 7,949 chunks (500/100-character chunking, 1,024-dim embeddings) in a FAISS vector store and externalizing all domain-specific configuration (sources, prompt, retrieval parameters) into a single reusable module.
- Implemented a document ingestion layer that extracts text with PyMuPDF and falls back to a Tesseract OCR sidecar for scanned books, streaming page renders through stdin at 250 DPI to avoid temporary files and keep interrupted runs resumable.
- Configured MMR retrieval (k=6, fetch_k=20, λ=0.5) to return passages that are both relevant and non-duplicate, and added a distance-threshold guard (0.72) that short-circuits out-of-scope queries before the LLM is invoked, reducing hallucination on noise input.

Cycle Catalog – Catalog, Orders & Administration Platform | *PHP, Laravel, Blade, MySQL*

- Collaborated in a four-person team to develop a full-stack catalog and order-management platform for a real client, supporting product reservations and orders without online payment processing.
- Implemented customer-facing catalog features, including product images, prices, descriptions, brands, search filters, featured-product carousels, customer sign-in, and product ratings.
- Developed administrative functionality for sales monitoring, product and supplier CRUD operations, and order notifications.
- Implemented data import/export workflows in XML, JSON, PDF, and CSV, with date-based filters, searchable tables, sequential identifiers, and printable invoices.

BusGo – Real-Time Bus Tracking Platform | *Java, Docker, PostgreSQL, Microservices, React Native*

- Developed a microservice-based bus tracking platform using Java, Docker, and PostgreSQL, with a React Native mobile interface.
- Implemented map-style route tracking and nearby-stop visualization based on a user-selectable distance range of 1–10 km.

EDUCATION

Universidad Nacional, Heredia, Costa Rica Completed 2026
Bachelor of Information Systems Engineering
Academic requirements completed in 2026; official degree conferral expected May 2027.
Relevant Coursework: Data Structures, Database Design and Implementation, Database Administration, Operating Systems, Mobile Platforms Programming, Systems Engineering Project

LANGUAGES

English: B2 / Upper-Intermediate | Spanish: Native